

Worksheet - 2

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Branch: MCA general

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Subject Name: Internet of Things

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Subject Code: 25CAH-708

Aim/Overview of the practical:

Interface a led, pushbutton (**Pull Up**) and LCD with Arduino Uno. Use two led and two pushbuttons. Use pushbutton as toggle switch.

1. Case 1: When press button 1 once then LED 1 will turn ON and print “**Button1 ON**” in Row 1 and “**LED1 ON**” in Row 2.
2. Case 2: When we press button 1 again then LED 1 will turn OFF and print “**Button1 OFF**” in Row 1 and “**LED1 OFF**” in Row 2.
3. Case 3: When we press button 2 once then LED 2 will turn ON and print “**Button2 ON**” in Row 1 and “**LED2 ON**” in Row 2.
4. Case 4: When we release button 2 again then LED 2 will turn OFF and print “**Button2 OFF**” in Row 1 and “**LED2 OFF**” in Row 2.

Apparatus:

1. Tinkercad
2. Arduino UNO R3
3. Breadboard Small
4. LED
5. Resistor
6. Push Button
7. LCD

Hardware Requirements:

1. Arduino UNO R3
2. Breadboard Small
3. 2 LEDs
4. 2 220 Ω Resistors
5. 2 10 K Ω Resistors
6. 2 Push Buttons
7. 1 LCD
8. Jumper Wires

Software requirements:

1. Tinkercad
2. Arduino IDE

Circuit Diagram and Code:

```
#include <Adafruit_LiquidCrystal.h>

Adafruit_LiquidCrystal lcd(0);

// Circuit pins const int

button1 = 4; const int button2

= 5; const int led1 = 10; const

int led2 = 11; // LED states

bool led1State = false; bool

led2State = false; // Previous

button states int lastButton1;

int lastButton2; void setup() {

    // Buttons

    pinMode(button1, INPUT);

    pinMode(button2, INPUT);

    // LEDs  pinMode(led1,

    OUTPUT); pinMode(led2,

    OUTPUT); // LEDs

    initially OFF

    digitalWrite(led1, LOW);

    digitalWrite(led2, LOW);

    // LCD  lcd.begin(16, 2);

    lcd.setBacklight(1);

    lcd.setCursor(0, 0);
```

```
lcd.print("System Ready");

lcd.setCursor(0, 1);

lcd.print("Press Button");

delay(1000);

// Read initial button state

lastButton1 = digitalRead(button1);

lastButton2 = digitalRead(button2);

lcd.clear();

}

void loop() {  int currentButton1 =

digitalRead(button1);  int currentButton2 =

digitalRead(button2);

// BUTTON 1  if (lastButton1 == HIGH &&

currentButton1 == LOW)

{  led1State = !led1State;

digitalWrite(led1, led1State);

lcd.clear();  if (led1State)  {

lcd.setCursor(0, 0);

lcd.print("Button1 ON");

lcd.setCursor(0, 1);

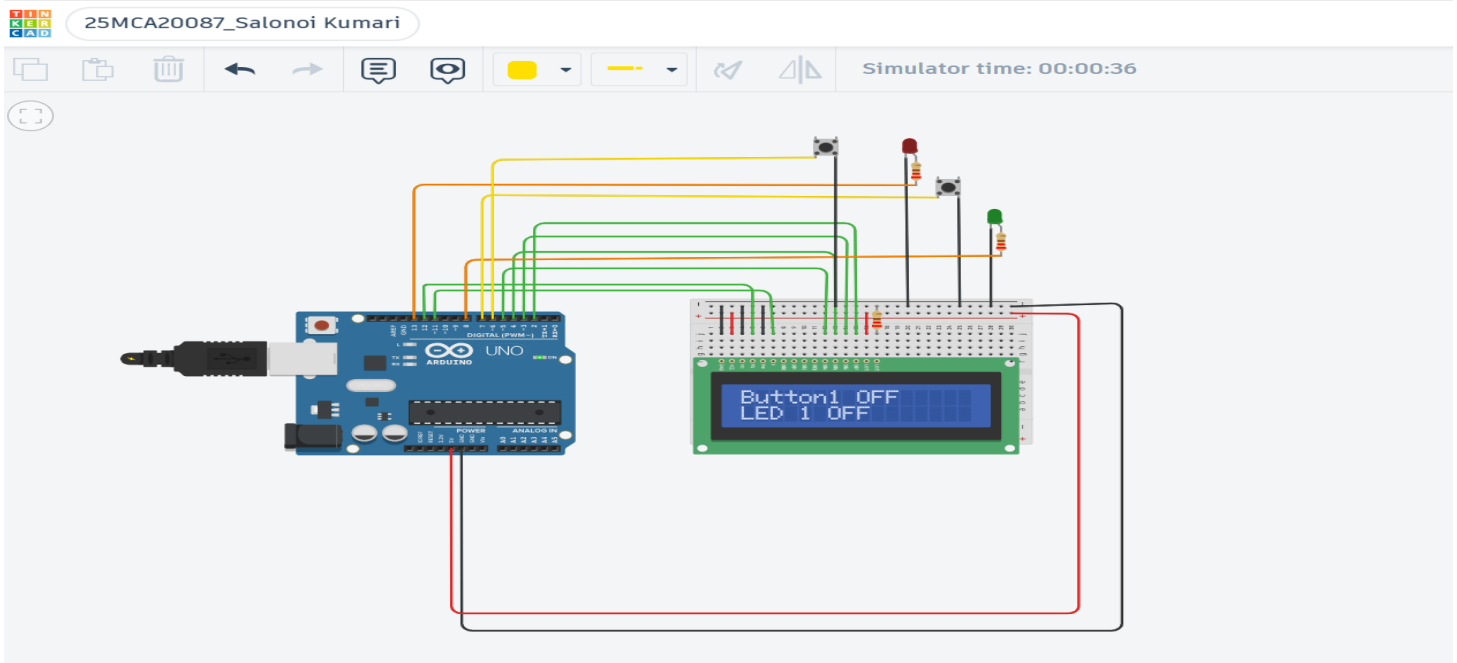
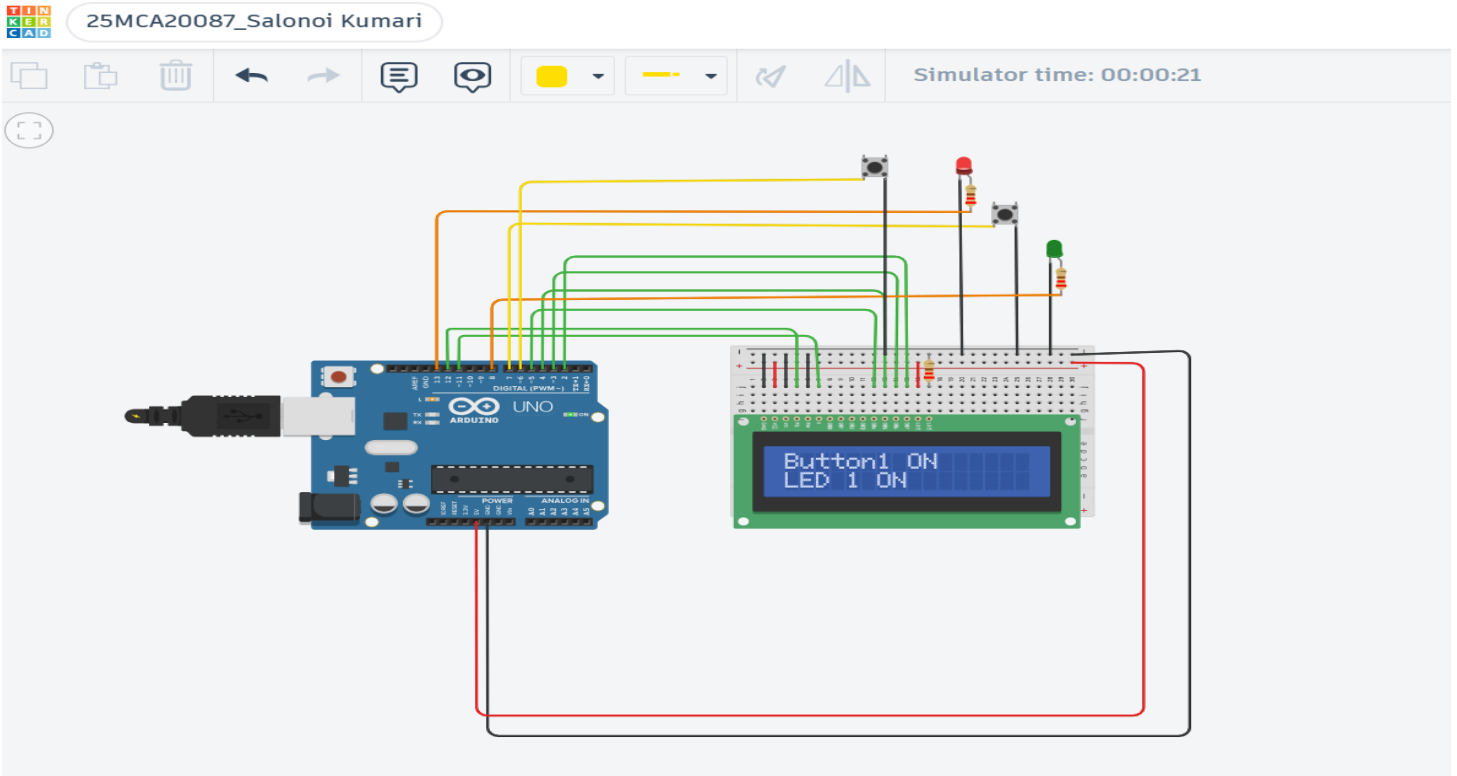
lcd.print("LED 1 ON");

}  else  {  lcd.setCursor(0, 0);

lcd.print("Button1 OFF");

lcd.setCursor(0, 1);
```

```
lcd.print("LED 1 OFF");
}
delay(250);
}
```



```
// BUTTON 2  if (lastButton2 == HIGH &&
currentButton2 == LOW)

{   led2State = !led2State;

digitalWrite(led2, led2State);

lcd.clear();  if (led2State)  {

lcd.setCursor(0, 0);

lcd.print("Button2 ON");

lcd.setCursor(0, 1);

lcd.print("LED 2 ON");

    }   else   {

lcd.setCursor(0, 0);

lcd.print("Button2 OFF");

lcd.setCursor(0, 1);

lcd.print("LED2 OFF");

    }

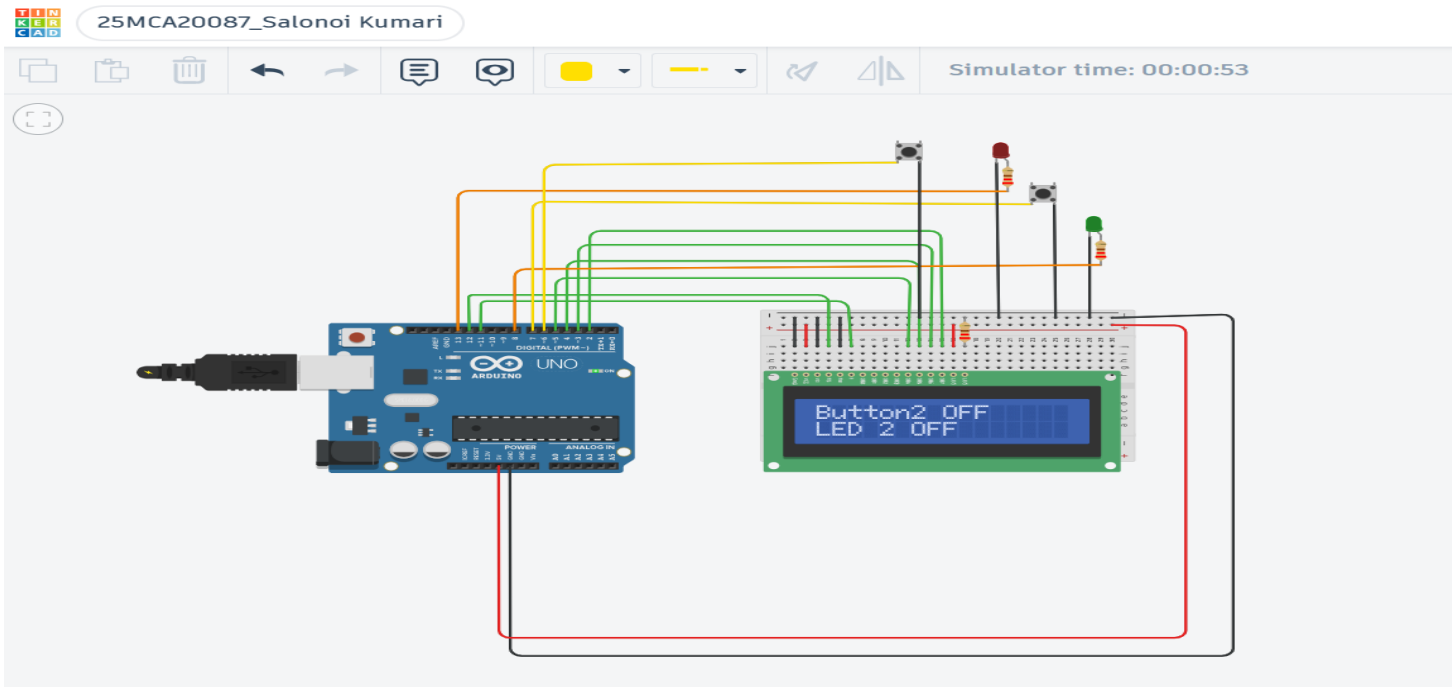
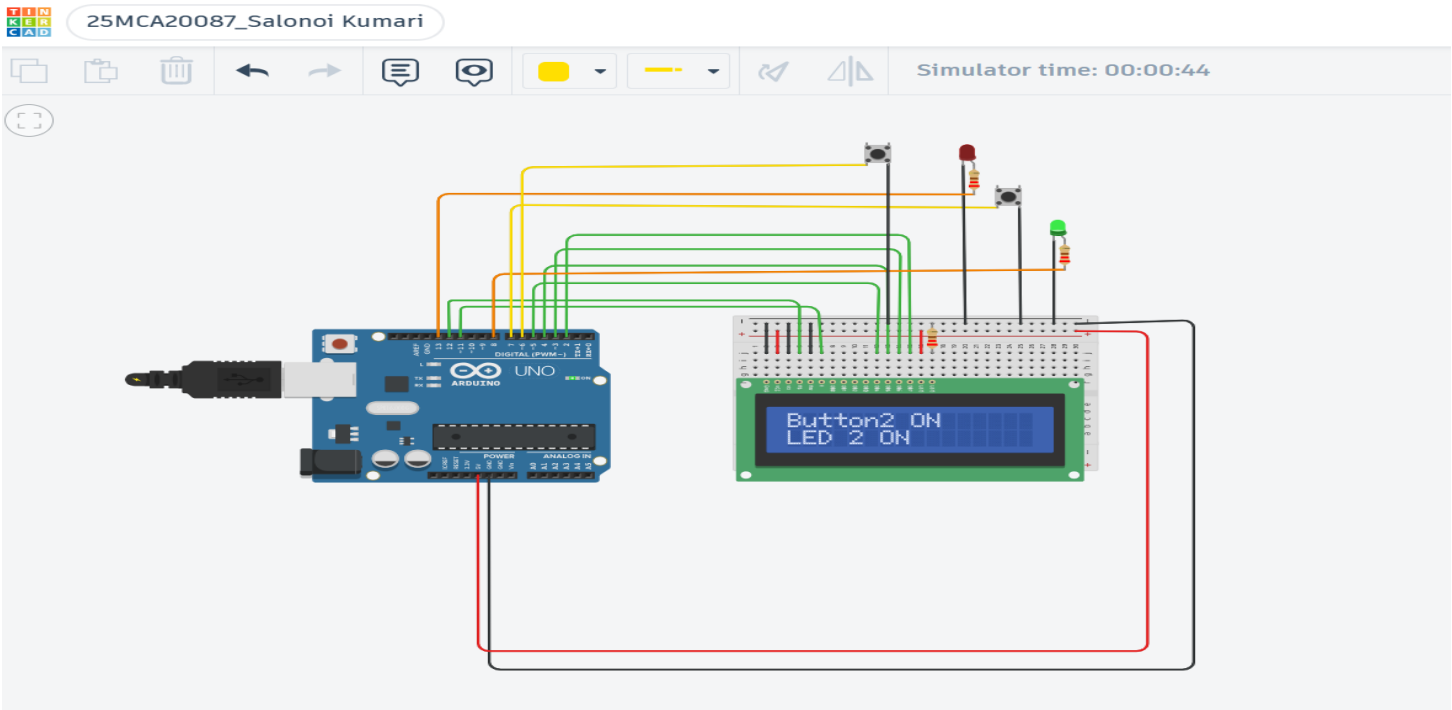
    delay(250);
}

// Save current button states

lastButton1 = currentButton1;

lastButton2 = currentButton2;
```

}



Learning outcomes (What I have learnt):

1. Learned about the **Arduino Uno**, its digital pins, and how it is used to control electronic components like LEDs, LCDs etc.
2. Understood the purpose of a **breadboard, LEDs, LCDs, Push Buttons, resistors, and jumper wires**, and how to connect them correctly to build a circuit.
3. Learned how to write and upload an **Arduino program** to control LEDs.
4. Gained practical knowledge of interfacing **LED and LCD** by Arduino.